

► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

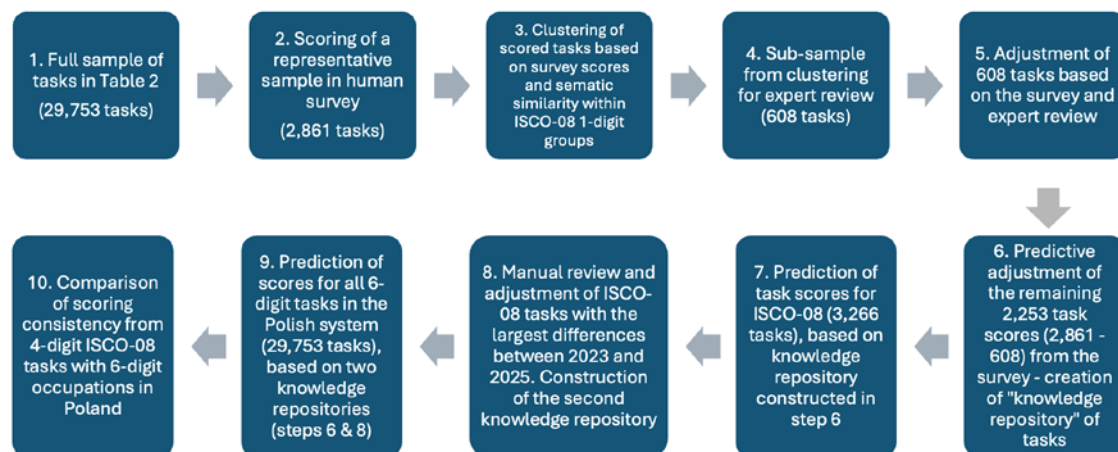
Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people

► 4 Expert validation survey

The expert group includes international specialists at the ILO and its partner organizations, NASK and the Polish Ministry of Family, Labour and Social Policy.⁹ The primary criterion for expert selection is their understanding of labour markets, including the complexity of occupations and tasks, and their real-life applications. While the experts are expected to have familiarity with the current family of generative AI technologies and their applications, expertise in AI design or development is not a requirement. This approach ensures that the focus remains on the feasibility and practical implications of automation within existing technological constraints, as technology experts tend to overstate technological capacities and risks when evaluating broader applications, particularly in workplace settings (Karger et al., 2023). This tendency is evident in earlier automation studies, which often assigned high displacement potential to routine tasks or entire occupations, such as garment production. As de Mattos et al. (2020) illustrate, these technological predictions frequently failed to account for the challenges of automating tasks requiring high dexterity or handling complex materials, such as stitching pliable fabrics. Accordingly, the assessment of automation potential in our study prioritizes expertise of task execution and workplace realities over purely technological expertise.

Since the experts cannot review all 2,861 tasks used in the survey, we focus on a representative sub-sample, selected in such a way to subsequently permit adjustments to entire task clusters. The overall scheme of this score adjustment procedure is presented in Figure 11.

► Figure 11. Schema of the scoring stages



First, we cluster the tasks into four fixed intervals, based on the distribution of the mean scores from the survey.¹⁰ We then use hierarchical clustering to group similar tasks within those clusters based on their semantic content. First, we assign embeddings to each task description using the *text-embedding-ada-002* model from OpenAI. Next, for each subgroup – defined by the combination of an occupational group at ISCO-08 1-digit level and a particular survey score cluster – we

⁹ A full list of contributors is provided in the acknowledgements section.

¹⁰ $\mu \in [0, 0.1] \cup (0.1, 0.3] \cup (0.3, 0.5] \cup (0.5, 1]$

► 7 Prediction of synthetic task-level scores for all ISCO-08 and 6-digit occupations

As the final step, we use the adjusted survey scores for 2,861 tasks to develop an AI assistant capable of predicting synthetic scores for new tasks across various classification systems. This combines the strengths of human expertise with generative AI's ability to generate consistent, large-scale responses based on precise instructions.

► Text Box 3. The conceptual structure of the LLM prompt (Python code for GPT-4o)

```
response = [...
    "role": "user",
    "content": (
        f"### TASK DETAILS\n"
        f"Job Task: {Tasks_STATS}\n"
        f"ISCO-08 4-digit code: {ISCO_08}\n"

        # f"### BACKGROUND\n"
        # "- In 2023, tasks were scored on their automation potential with Generative AI technology, by
asking GPT-4 model
        to provide a score."
        # "- The scores had a scale of 0 to 1, where 0 indicates no potential for automation using Generative
AI, and 1
        indicates full potential (no human operator needed).\n"
        # f"The score for this task assigned in the 2023 study was: {score_2023}\n\n"

        "### CONTEXT\n"
        "A large-scale survey was conducted with participants from diverse professional and technological
backgrounds.
        These participants:\n"
        "- Had practical knowledge of the work and tasks being evaluated and a theoretical knowledge of
labour markets.\n"
        "- Scored tasks on a scale of 0 to 1, where 0 indicates no potential for automation using Generative
AI, and 1
        indicates full potential (no human operator needed).\n"
        f"The 50 tasks with lowest scores were: {bottom50_scores_dict}"
        f"The 50 tasks with highest scores were: {top50_scores_dict}"
        f"This particular task was not scored, but 30 tasks with semantic proximity to this task received the
following scores:
        {context_query}"

        f"### YOUR TASK\n"
        "Predict the adjusted score for this task, considering:\n"
        "- The lowest and highest scores in the context you were provided.\n"
        "- Adjusted scores of semantically similar tasks in the context you were provided.\n"
        "- The capabilities of Generative AI to automate the task.\n"
        "- The assumption that the task is performed in a high-income country like Poland, where access to
computers and the
        internet is common.\n\n"

        f"### IMPORTANT INSTRUCTIONS\n"
        "- Do not simply average the scores; analyze the task and adjust accordingly.\n"
        "- Provide only the score in line 1 and your justification in next lines. Do not provide any additional
text or
        commentary.\n"
        "- Do not provide a range; give a single score for the task."))]
```

To develop our predictive model, we evaluated two approaches: (i) fine-tuning a large language model (LLM) with the 2,861-task dataset or (ii) designing an advanced prompting strategy (prompt engineering). Fine-tuning, while promising for learning domain-specific patterns, is